

1984/1. Prove that

$$0 \leq yz + zx + xy - 2xyz \leq \frac{7}{27}$$

where x, y, z are non-negative real numbers for which

$$x + y + z = 1$$

1984/2. Find one pair of positive integers a and b such that:

(i)

$$ab(a + b)$$

is not divisible by 7;

(ii)

$$(a + b)^7 - a^7 - b^7$$

is divisible by 7^7 .

Justify your answer.

1984/3. In the plane two different points O and A are given.

For each point X of the plane, denote by $\alpha(X)$ the measure of the angle between OA and OX in radians, counterclockwise from OA

$$0 \leq \alpha(X) < 2\pi$$

Let $C(X)$ be the circle with center O and radius of length

$$OX + \frac{\alpha(X)}{OX}$$

Each point of the plane is colored by one of a finite number of colors.

Prove that there exists a point Y for which $\alpha(Y) > 0$ such that its color appears on the circumference of the circle $C(Y)$.

1984/4. Let $ABCD$ be a convex quadrilateral such that the line CD is a tangent to the circle on AB as diameter.

Prove that the line AB is a tangent to the circle on CD as diameter if and only if the lines BC and AD are parallel.

1984/5. Let d be the sum of the lengths of all the diagonals of a plane convex polygon with n vertices ($n > 3$), and let p be its perimeter.

Prove that

$$n - 3 < \frac{2d}{p} < \left[\frac{n}{2} \right] \left[\frac{n+1}{2} \right] - 2$$

where $[x]$ denotes the greatest integer not exceeding x .

1984/6. Let a, b, c and d be odd integers such that

$$0 < a < b < c < d$$

and

$$ad = bc$$

Prove that if

$$a + d = 2^k$$

and

$$b + c = 2^m$$

for some integers k and m , then

$$a = 1$$